

Surya Attendance & Payroll Management System

End-to-End Implementation Report & Architecture Specification

1. Executive Summary

The **Surya Attendance & Payroll Management System** is a lightweight, secure Single Page Application (SPA) designed for employee time-tracking, shift schedules, geo-verification, and payroll processing. Built for **HS Group Delhi (House of Surya)**, the application facilitates employee check-ins, shift swap request coordination, profile database management, and payroll statements, with strong role-based access control (RBAC).

2. System Architecture

The application runs as a front-end SPA supported by a mocked database layer stored in browser `localStorage``.

- **Routing Layer (`js/app.js`)**: A client-side hash router that parses navigation links (e.g., ``#dashboard``, ``#admin-schedules``, ``#employee-swaps``) and mounts views dynamically.
- **Database & Calculation Layer (`js/db.js`)**: An object-oriented API that performs schema queries, database transactions, shift assignments, and payroll calculations.
- **Security & Authentication (`js/auth.js`)**: Validates passwords, tracks active sessions, and simulates biometric registers.

3. Phased Implementation Details

Phase 1: Location & Geofencing System Overhaul

- **Mock Office Geofence Coordinates:**
 1. **Kohat Enclave, Pitampura, Delhi (HQ)** — Coord: ``28.6978° N, 77.1408° E`` (In-Range).
 2. **Chandni Chowk (Hub)** — Coord: ``28.6562° N, 77.2310° E`` (Out of Range - 9.8 km deviation).
 3. **Omaxe City, Delhi (Branch)** — Coord: ``28.8130° N, 77.0673° E`` (Out of Range - 14.5 km deviation).
- **Clock-in Verification Logic:**
 - Biometric actions (Face ID, Fingerprint) are restricted to the HQ location.
 - Attempting to clock in from an out-of-range location disables biometric buttons.
 - Employees can check in via Passcode only after filling out an **Out-of-Geofence Deviation Justification** textarea.

- Out-of-range logs are flagged as `Pending Manager Review`.

Phase 2: Shift Schedule Location Selector

- **Shift Settings Modal:** A location dropdown selection was added to the **Create / Edit Shift Calendar** modal.
- **Inline Shift Card Controls:** Added interactive **Location Select** and **Location Swap** dropdown selectors directly onto the shift cards on the admin page.
- Changing location instantly saves modifications to `localStorage`.
- Selecting another shift from the swap dropdown swaps the locations between both patterns immediately.

Phase 3: Location-Aware Shift Swapping

- **Coworker Preview:** The coworker swap dropdown displays each coworker's assigned shift name and location.
- **Coworker Assignment Preview Block:** Selecting a coworker displays a detailed current assignment card showing their shift times and location before submission.
- **Swap Mode Configurations:**
 1. **Swap Both Shift & Work Location:** Exchanges the shift timings and the preferred work locations of both users.
 2. **Swap Work Location Only:** Exchanges the preferred work locations while keeping their current shift timings.
 3. **Swap Shift Only:** Exchanges shift timings while keeping their current preferred work location.
- **Ledgers Update:** Updated the Sent Requests, Received Requests, and Manager Approval logs to display locations and swap modes.

Phase 4: Role-Based Payroll & Profile Access

- **Segregation of Duties:** Both the **HR Coordinator** and **Operations Manager** can now access, edit, and configure the payroll/salaries of general Employees (role === 'employee').
- **HR Payroll Controlled by Manager:**
 - Operations Manager can view, edit, and delete the HR Coordinator's profile and payroll.
 - The HR Coordinator is hidden from the register table when logged in as HR, preventing them from modifying their own salary, base payroll coefficients, or portal access roles.
- **Role Escalation Protection:** The user editor modal restricts portal role assignments; only the Operations Manager can change user access roles (assigning `Employee`, `HR Coordinator`, or `Operations Manager`). When edited by HR, the selection is restricted to the `Employee` role.

Phase 5: Manager Payslip Adjustments

- **Interactive Adjustments Drawer:** Added an **Edit Adjustments** button to the Admin Payslip Preview drawer (visible only to the logged-in Operations Manager).
- **Adjustment Modals:** Clicking this button opens a modal to customize:
 - **Discretionary Bonus** (Earning Addition)

- **Ad-hoc Deductions** (Deduction Subtraction)
 - **Manager Remarks / Notes** (Printed directly onto the payslip statement)
 - **Calculations & Persistence:** Integrated adjustments directly into the database calculations (`js/db.js`). The net disbursed payouts reflect the overridden values immediately.
 - **Payslip Template Update:** Modified both the administrator inspect template and the employee's personal payslip statement (`js/app.js`) to display discretionary bonuses, custom deductions, and remarks dynamically when present.
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4. Verification & Testing Guide

Running the App Locally

The application is served locally via a Python HTTP server on port 8012:

```
```bash
python -m http.server 8012
```
```

To run the server, browse to: **[<http://localhost:8012>](http://localhost:8012)**

Simulated User Credentials

1. **Operations Manager** (Full Control, edit HR, adjust payslips):
 - Username: `manager`
 - Password: `ManagerPassword123!`
2. **HR Coordinator** (Manage employee shifts & profiles, no self-edits):
 - Username: `hr`
 - Password: `HRPassword123!`
3. **General Employee** (John Doe):
 - Username: `john`
 - Password: `JohnPassword123!`